Open problems:
Schrödinger equation is equivalent to information conservation

Open problems raised by the technical brief in this directory, and by its Lean formalization (see `formalization-log.tex`).

From the brief

1. **Classical counterpart.** State and prove the classical analogue: Hamiltonian evolution is equivalent to preservation of the Shannon entropy and of "the definitional independence of degrees of freedom". (Brief, Sec. 4: "This parallels the classical case where Hamiltonian evolution is equivalent to the preservation of Shannon entropy (Liouville's theorem)".)

2. **Entropy conservation as determinism and reversibility.** The brief reads DR-INFO as "an information theoretic way to characterize determinism and reversibility". A precise definition of deterministic and reversible evolution, and a proof that it is equivalent to DR-INFO, are not given. (Brief, Sec. 1.)

3. **Propagator for unbounded time-dependent $H(t)$.** "The Schrödinger equation gives rise to a differentiable unitary propagator $U_{t_0 \to t_1}$" is assumed. For a continuous family of bounded self-adjoint $H(t)$ this is now proved, in any dimension, from the global existence and uniqueness theorem for linear ODEs: the propagator exists, is unique and is unitary (`exists_propagator_of_continuous`, `propagator_unique`), and in finite dimension its two-parameter evolution satisfies [B]EVMIX, DR-SCEQ, DR-INFO and EVDIFF (`TwoParameter.exists_propagator_isEvolution`; log, "The propagator of a time-dependent Hamiltonian"). Existence, uniqueness and unitarity also hold for bounded $H(t)$ that are only strongly continuous, with $t \mapsto U(t)\psi$ differentiable for every ψ (`exists_propagator_of_continuous_apply`), and so does EVDIFF: $U(t)\rho U(t)^\dagger$ is differentiable in operator norm for every density operator ρ (`hasDerivAt_conj_of_continuous_apply`). The propagator itself need not be norm-differentiable: on $L^2[0,1]$, multiplication by the indicator of $[0,t]$ is such an $H(t)$, and no propagator of it is norm-differentiable on $(0,1)$ (`indicatorHamiltonian_counterexample`). For unbounded self-adjoint $H(t)$ existence needs Kato-type hypotheses and remains open. The converse has the same gap: for unbounded $H(t)$, Step 3 gives only a symmetric $H(t_0)$ (next section, first item). (Brief, Sec. 3, proof of Theorem 2 ($\Rightarrow$); Log, "Differences from the brief".)

From the formalization

Beyond finite dimension the formalization has a part in any dimension; it leaves open the items "Unbounded time-dependent Hamiltonians", "Equation at every time in infinite dimension" and "Wold–Cooper reduction" below. Whether a phase choice makes $t \mapsto U(t)$ differentiable in operator

norm in infinite dimension is settled negatively: on $\ell^2(\mathbb{N})$, a phase e^{it} on the basis vector $e_{\lceil 1/|t| \rceil + 1}$ satisfies all hypotheses, with EVDIFF in operator norm, but no unitary family inducing it is norm-differentiable at 0 (`movingPhase_counterexample`; log, "No norm-differentiable propagator").

1. **Unbounded time-dependent Hamiltonians.** EVDIFF on every density operator forces a bounded H (Banach–Steinhaus). With EVDIFF on a dense set of states, Theorem 2 is formalized for a time-independent evolution, with unbounded self-adjoint H (`drSceq_iff_drInfo_and_evDiff`, by Stone's theorem). Without EVTI, Step 3 gives only a skew-symmetric $A(t_0)$, and $H(t_0)$ need not be self-adjoint: the propagator of $H(t) = H_\beta$ for $t < 0$ and H_α for $t \geq 0$, two self-adjoint extensions of the momentum on $L^2(0,1)$, is differentiable at 0 exactly on the domain of the minimal momentum operator, which is not self-adjoint. Open: the hypotheses (Kato-type, or on the evolution) under which DR-INFO + EVDIFF gives a self-adjoint $H(t)$. The example is formalized: the vectors on which the propagator is differentiable at 0 form a dense subspace, on which $i\hbar U'(0)$ is symmetric but not self-adjoint (`switchedHamiltonian_counterexample`). This subspace is $H_0^1(0,1)$, the domain of the minimal momentum, and $i\hbar U'(0)$ is $-i\, d/dx$ on it (`switchedDomain_eq`). The two pieces of U are $e^{-itH_\alpha/\hbar}$ and $e^{-itH_\beta/\hbar}$ (`twistedTranslation_eq_schrodingerPropagator`). (Log, "Unbounded Hamiltonians".)

2. **Equation at every time in infinite dimension.** In finite dimension, one differentiable unitary $U(t)$ for all times solves the Schrödinger equation at every time, and $\Phi_{t_0 \to t_1}$ is conjugation by $U(t_1)U(t_0)^\dagger$ (`TwoParameter.drSceq_iff_drInfo_and_evDiff`). The proof fixes the determinant of the local unitaries, which leaves only a root of unity to match when gluing them. In any dimension the determinant is not defined, and the local unitaries are only strongly differentiable, and in general cannot be chosen norm-differentiable (`movingPhase_counterexample`), so the global propagator is not built there. (Log, "The equation at every time".)

3. **Wold–Cooper reduction.** The right shifts V_t on $L^2(0,\infty)$ give an evolution that satisfies [B]EVMIX, maps pure states to pure states, preserves transition probabilities and the entropy, but is induced by no unitary; this is formalized (`shift_semigroup_counterexample`, in the brief "Multiple equivalent characterizations of the Schrödinger equation"). The Wold–Cooper decomposition, which reduces every strongly continuous semigroup of isometries to a unitary group plus copies of this one, is not formalized. (Log, any-dimension part, "Problems found in the brief", second item.)